This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset LaTeX solutions.

1.a

1.b **Answer:** $O(mnp)$.

Justification. The product $C = AB$ has $m \cdot p$ entries, and each entry is an inner product of length n:

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj},$$

so each entry costs n multiply–adds. The total is therefore $mp \cdot n = O(mnp)$.

Two caveats. This is the *standard* (naive) algorithm — Strassen-style methods beat $O(n^3)$ on square matrices. And as a sanity check, the square case $m = n = p$ gives $O(n^3)$, as expected.

1.c

1.d **(i)** Xw**, shapes** $(n, d), (d,) \to (n,)$

```
einsum(X, w, 'n d, d -> n')
```

The single axis of w is labelled `d` so that it aligns with the column axis of X. Since `d` is absent from the output it is summed over, giving $(Xw)_i = \sum_d X_{id} w_d$. The axis `n` is carried through, so the result has one entry per row of X.

(ii) $XX^\top$**, shapes** $(n, d), (n, d) \to (n, n)$

```
einsum(X, X, 'i d, j d -> i j')
```

Entry $[i, j]$ is the dot product of row i with row j, that is $(XX^\top)_{ij} = \sum_d X_{id} X_{jd}$. The two row axes index *different* rows, so they must receive different letters `i` and `j`, both of which are kept in the output. The feature axis `d` is shared by both operands and dropped from the output, so it is contracted. No explicit transpose is required — the transpose is expressed by *which letters line up*, not by flipping an array.

(iii) $\mathrm{diag}(X^\top X)$**, shapes** $(n, d), (n, d) \to (d,)$

```
einsum(X, X, 'n d, n d -> d')
```

The k-th diagonal entry of $X^\top X$ is the squared norm of column k: $\mathrm{diag}(X^\top X)_d = \sum_n X_{nd} X_{nd} = \sum_n X_{nd}^2$. Both operands are the plain array X, so both are labelled `n d`. The row axis `n` is absent from the output and so is summed; the feature axis `d` appears in both inputs *and* the output, so it is kept and aligned elementwise. The net effect is to square X elementwise and then sum down the rows.

Note that $X^\top X$ is itself $d \times d$; it is diag that collapses it to a length-d *vector*. Computing it as `np.diag(X.T @ X)` would build the full $d \times d$ matrix only to discard the off-diagonal entries — $O(nd^2)$ work for d useful numbers — whereas the einsum string never forms that matrix and costs $O(nd)$. The cost can be read straight off the letters, as in 1b: `'n d, n d -> d'` has only two distinct letters, hence two loops.

2.a
2.c

3.a
3.b

4.a
4.b
4.c
4.d

5.a
5.b
5.c
5.d